SAI TEJA PEGALLAPATI

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Data Science graduate with strong experience in Big Data analysis, Machine Learning, and Data Mining. Data Engineer and DevOps professional with 2+ years of experience in cloud computing, data pipeline development, and enterprise system integration. AWS Certified Solutions Architect passionate about delivering scalable and efficient data solutions.

SKILLS

Programming: Python, C, Java, HTML & CSS, JavaScript
Machine Learning: Scikit-learn, Pandas, NumPy, Matplotlib, PyTorch, TensorFlow/Keras, scikit-learn
Big Data: Hadoop, Spark
DevOps: Azure DevOps, Git, Bash Scripting, LINUX Scripting
Databases: MySQL
Middleware: TIBCO EMS, TIBCO BW6/BW5, TIBCO Administrator, TIBCO Hawk
ITSM: ServiceNow
Tools: Tableau, Power BI, Excel, VS Code, MATLAB

Certifications: AWS Certified Solutions Architect – Associate (Issued: May 2023)

EDUCATION

University of Houston Aug 2024 – Present

M.S. in Engineering Data Science *GPA: 3.67/4.0*

Coursework: Introduction to Machine Learning, Artificial Neural Network (ANN), Cloud Computing, Data Mining, Big Data Analytics, Data Science, Probability and Statistics

Anurag Groups of Institutions (CVSR) Aug 2017 – Aug 2021

B.E. in Electronics and Communication Engineering *GPA: 8.99/10.0*

Coursework: Object-oriented Programming(JAVA), Digital Signal Processing, Digital Image Processing, Micro Processors and Micro Controls, Embedded systems, VLSI Designing, Optical & Satellite communication, Linear Algebra, Advanced Calculus

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) — System Engineer Oct 2021 — Jan 2024

- Designed and managed scalable TIBCO-based enterprise integrations across Dev, QA, UAT, and Prod environments.
- Built CI/CD pipelines using Azure DevOps, reducing deployment time by 30% and improving release consistency.
- Automated deployments with CLI and ANT scripts; integrated systems via REST, SOAP, JDBC, and JMS.
- Migrated TIBCO applications to Docker containers to enhance scalability and portability.
- Implemented infrastructure monitoring using TIBCO Hawk, Splunk, and custom log parsers.
- Boosted system reliability by 20% through proactive configuration management.
- Collaborated with cross-functional teams to enforce ITIL practices and ensure operational excellence.

PROJECTS

Anomaly Detection in Network Traffic (Deep Learning Project) March 2025- May 2025
University of Houston | Python, PyOD, AutoEncoders, PCA, KDD Cup 1999 Dataset

- Applied unsupervised learning to detect cybersecurity anomalies using classical (KNN, Isolation Forest) and deep learning (AutoEncoder, Variational AutoEncoder) models.
- Achieved >95% accuracy and high recall (0.97) with AutoEncoder-based models trained exclusively on normal data.
- Reduced dimensionality using PCA, visualized results via t-SNE and ROC curves, and validated model thresholds

based on reconstruction error.

- Showcased strong understanding of high-dimensional data preprocessing, class imbalance handling, and anomaly detection pipelines.

Concrete Strength Prediction Using Deep Neural Networks Feb 2025 – May 2025
University of Houston | TensorFlow/Keras, Regression Modeling, Materials Data

- Designed and trained a multi-layer feedforward neural network to predict concrete compressive strength from 8 input features.
- Achieved **low** MAE and RMSE, outperforming linear regression baselines by capturing non-linear feature interactions.
- Incorporated BatchNormalization, Dropout, and dynamic learning rate adjustment, ensuring robust generalization and training efficiency.
- Demonstrated practical application of deep learning in engineering/material science, from EDA to deployment-ready modeling.